

$$[4.6] \quad (1) \quad X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X-\mu}{\sigma} \sim N(0,1) \quad \text{f}$$

$$\begin{aligned} & P(\mu + a\sigma \leq X \leq \mu + b\sigma) \\ &= P\left(\frac{(\mu+a\sigma)-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{(\mu+b\sigma)-\mu}{\sigma}\right) \\ &= P(a \leq Z \leq b) \end{aligned}$$

$$\begin{aligned} (a) \quad P(\mu - \sigma \leq X \leq \mu + \sigma) &= P(|Z| \leq 1) \\ &= 2 \times 0.8413 - 1 \\ &= 0.6826 \end{aligned}$$

$$\begin{aligned} (b) \quad P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) &= P(|Z| \leq 2) \\ &= 2 \times 0.9772 - 1 \\ &= 0.9544 \end{aligned}$$

$$\begin{aligned} (c) \quad P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) &= P(|Z| \leq 3) \\ &= 2 \times 0.9987 - 1 \\ &= 0.9974 \end{aligned}$$

(2) 3

(3) (a) 2.327

(b) -2.327

(c) 2.575

$$[4.7] \quad (1) \quad \text{標準正規分布 } f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad x \in \mathbb{R}$$

$$\begin{aligned} f(-x) &= \frac{1}{\sqrt{2\pi}} e^{-\frac{(-x)^2}{2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = f(x) \end{aligned}$$

正規分布の性質

$$\begin{aligned} P(|Z| \geq a) &= P(Z < -a, a < Z) \\ &= \int_{-\infty}^{-a} f(x) dx + \int_a^{\infty} f(x) dx \quad \downarrow u = -x \\ &= \int_{\infty}^a -f(-u) du + \int_a^{\infty} f(u) du \\ &= \int_a^{\infty} f(u) du + \int_a^{\infty} f(u) du \\ &= 2 \int_a^{\infty} f(u) du = 2P(Z \geq a) \end{aligned}$$

正規分布

$$\begin{aligned} (2) \quad P(|X-\mu| \geq \frac{\sigma}{4}) &= 2P(X-\mu \geq \frac{\sigma}{4}) \\ &= 2P\left(\frac{X-\mu}{\sigma} \geq 0.25\right) \\ &= 2 \times (1 - 0.5987) \\ &= 0.8026 \\ &\approx 0.803 \end{aligned}$$

$$\begin{aligned} (3) \quad E[\bar{X}] &= \frac{n}{n} \times \mu = \mu \\ V[\bar{X}] &= \frac{\sigma^2}{n^2} \times n = \frac{\sigma^2}{n} \\ \bar{X} &\sim N\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1) \\ \bar{X}-\mu &= \frac{\sigma}{\sqrt{n}} Z \quad \text{f} \\ P(|X-\mu| \geq \frac{\sigma}{4}) &= P\left(\frac{\sigma}{\sqrt{n}} |Z| \geq \frac{\sigma}{4}\right) \\ &= P\left(|Z| \geq \frac{\sqrt{n}}{4}\right) \leq 0.02 \\ &= 2 - 2P\left(Z \leq \frac{\sqrt{n}}{4}\right) \approx 0.02 \end{aligned}$$

$$\begin{aligned} P\left(Z \leq \frac{\sqrt{n}}{4}\right) &\leq 0.99 \\ 2.33 &\leq Z \leq \frac{\sqrt{n}}{4} \\ n &\geq 5.4289 \times 16 \\ &\geq 86.8624 \quad \therefore n = 87 \end{aligned}$$

$$[4.8] \quad \mu = 66.31, \sigma = 23.55, n = 350198 \quad \text{149}$$

$$\begin{aligned} (1) \quad P(80 \leq X \leq 95) &= P\left(\frac{80-66.31}{23.55} \leq Z \leq \frac{95-66.31}{23.55}\right) \\ &= P\left(\frac{13.69}{23.55} \leq Z \leq \frac{28.69}{23.55}\right) \\ &\approx P(0.58 \leq Z \leq 1.22) \\ &= 0.8888 - 0.7190 \\ &= 0.1698 \end{aligned}$$

$$350198 \times 0.1698 \approx 59463$$

$$\begin{aligned} (2) \quad P(Z \leq 0.84) &\approx 0.8 \quad \text{f} \\ P(Z \leq 0.84) &= P(X \leq 66.31 + 23.55 \times 0.84) \\ &= P(X \leq 66.092) \approx 0.8 \end{aligned}$$

中心値より86点

$$[4.9] \quad (1) \quad \binom{6}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^3 = \frac{20}{2^6} = \frac{5}{16}$$

(2) 表が出る回数 X は $\text{Bin}(10000, 1/2)$ に従う

$$\begin{aligned} E(X) &= 10000 \times \frac{1}{2} = 5000 \\ \sqrt{V(X)} &= \sqrt{10000 \times \frac{1}{2} \times \frac{1}{2}} = \frac{100}{2} = 50 \end{aligned}$$

(3) $np = n(1-p) = 5000 > 5$ f, $\therefore$ 正規分布に近づく

$$\begin{aligned} P(4900 \leq X \leq 5100) &\quad \downarrow \text{連続補正} \\ &\approx P(4899.5 < X < 5100.5) \\ &= P\left(\frac{4899.5-5000}{50} \leq Z \leq \frac{5100.5-5000}{50}\right) \\ &= P(-2.01 \leq Z \leq 2.01) \\ &= P(|Z| \leq 2.01) \\ &= 2 \times (0.9791 - 0.5) \\ &= 0.9582 \end{aligned}$$